4.4.2 Customize CDN Caching Behavior

A CDN works by caching content. If the CDN has stale or bad content, then it is hurting rather than helping your application. The purpose of this section is to describe general caching behavior of most CDNs, your specific provider may behave slightly differently.

4.4.2.1 CDN Request Caching

While a CDN is described as being good for caching assets, in reality caches the entire request. This includes the body of the asset as well as any headers. The most important one being `Cache-Control`which tells the CDN (and web browsers) how to cache contents. This means that if someone requests an asset that does not exist `/assets/i-dont-exist.png` and your Rails application returns a 404, then your CDN will likely cache the 404 page if a valid `Cache-Control` header is present.

4.4.2.2 CDN Header Debugging

One way to check the headers are cached properly in your CDN is by using curl. You can request the headers from both your server and your CDN to verify they are the same:

```
$ curl -I http://www.example/assets/application-
d0e099e021c95eb0de3615fd1d8c4d83.css
HTTP/1.1 200 OK
Server: Cowboy
Date: Sun, 24 Aug 2014 20:27:50 GMT
Connection: keep-alive
Last-Modified: Thu, 08 May 2014 01:24:14 GMT
Content-Type: text/css
Cache-Control: public, max-age=2592000
Content-Length: 126560
Via: 1.1 vegur
```

Versus the CDN copy.

```
$ curl -I http://mycdnsubdomain.fictional-cdn.com/application-
d0e099e021c95eb0de3615fd1d8c4d83.css
HTTP/1.1 200 OK Server: Cowboy Last-
Modified: Thu, 08 May 2014 01:24:14 GMT Content-Type: text/css
Cache-Control:
public, max-age=2592000
Via: 1.1 vegur
Content-Length: 126560
Accept-Ranges:
bytes
Date: Sun, 24 Aug 2014 20:28:45 GMT
Via: 1.1 varnish
Age: 885814
Connection: keep-alive
X-Served-By: cache-dfw1828-DFW
X-Cache: HIT
X-Cache-Hits:
68
X-Timer: S1408912125.211638212,VS0,VE0
```

Check your CDN documentation for any additional information they may provide such as `X-Cache` or for any additional headers they may add.

4.4.2.3 CDNs and the Cache-Control Header

The cache control header is a W3C specification that describes how a request can be cached. When no CDN is used, a browser will use this information to cache contents. This is very helpful for assets that are not modified so that a browser does not need to re-download a website's CSS or javascript on every request. Generally we want our Rails server to tell our CDN (and browser) that the asset is "public", that